

Meeting with FWMai @June 30, 2025

Summary

This meeting focused on a project to optimize the PURE (Protein synthesis Using Recombinant Elements) system using AI to tune component concentrations for improved protein yield.

PURE System Overview

- PURE stands for Protein synthesis Using Recombinant Elements
- Focuses on transcription and translation to synthesize proteins
- Contains 48 components (technically around 60 when counting all amino acids)
- Main limitation: low yield compared to other systems
- Goal: improve yield through component optimization

Component Organization

- NEB Pure Express uses two mixtures:
 - Solution A (energy mix): SMS (small molecule solution) and tRNAs
 - Solution B: ribosomes and protein mix
- Project approach: separate SMS and tRNAs, and separate ribosomes and protein mix
- Components are grouped by function with dependencies (e.g., elongation factors)

Optimization Constraints

- Volume restrictions will remain constant initially
- Will adjust concentrations of reagents within mixtures

- Protein mix constraints:
 - No protein concentration should be zero
 - Overall protein concentration should be 8-25 mg/ml (soft constraint)
- Some components (particularly salts) have been previously optimized

Project Approach

- Initial plan: gather baseline data, then let AI suggest optimization
- Will include expert knowledge but avoid over-biasing the model
- When suggestions seem unlikely to work, team will either:
 - Run the experiment anyway to provide learning data
 - Classify as failure without running and retrain model

Timeline and Expectations

- Baseline data expected by end of the week
- Begin testing AI suggestions next week
- Goal is 5x improvement (reasonable based on previous studies)
- Long-term strategic project with target for significant progress by September conferences

Action Items

- ☐ Create list of proteins to focus optimization on first
- ☐ Prepare protein mix by the end of the week
- ☐ Run baseline experiments based on published paper
- ☐ Move materials to shared drive for team access
- ☐ Set up standard curve for consistent measurements
- ☐ Determine optimal number of iterations to run per week

Measurement Approach

- Currently using relative fluorescence with a positive control

- Consider using a standard curve to reduce variance
- May add PLAM as additional measurement

Notes

Transcript

Boom. Okay. So, from what we laid out... From what we spoke of last week, I thought we should just like... Create the template for y'all, put all the components on it, and then also just give you a quick, more in-depth overview of Pure. So what is Pure?

It's like the acronym actually stands for protein synthesis using recombinant elements. And then so it basically focuses on two functions transcription and translation in order to synthesize proteins. There are 48 components in Pure. Technically, there's two mixtures, the amino acids and the tRNA synthetases.

Oh, sorry. Is that right? It's not tyrannous. It's amino acid. Anyways, amino acids we're going to treat as one component, even though there's 20... 20 amino acids. That's why it's like 48. Technically, I think it's like 60 and so on. And so basically, one of the biggest limitations up here is that the yield is low compared to other systems.

So our goal with Find What Matters is basically to improve the yield of pure by tuning each component. OK, so how do we assemble these reactions? So putting 60 components together is, or how many components? 48 components together is too difficult, so we use mixtures. So NEB Pure Express, they have two mixtures that they use, which is solution A, which is energy mix.

And that contains SMS, which is a small molecule solution, and tRNAs together. And then they also have solution B, which has ribosomes and protein mix. And then you also have to add RNAs inhibitor, DNA of choice, and then nuclease-free water. In order to have a total volume 10, those three volumes are going to be constant. And then how are we different from Pure Express? So basically, we just separate the SMS and tRNAs for energy mix, and then we separate the ribosomes and protein mix.

from Solution B.

So why am I telling you this information? So we're going to keep the volumes constant for each component. And that will help our calculations a little bit easier. But I feel like once we get the baseline, then we can start changing the volumes of each component, if that makes sense.

Okay. And then I kind of try to divide the components based on function with the help of chatGBT. And then which was kind of nice. So it kind of just shows you like all the different components in there and then like what functions that provide. And then kind of like the dependencies.

So, like, for example, like, for translation – oh, is my mouse moving? No, it's not. But for translation factors in the protein mix... You know, there's like EF-TU, EF-TS, and EF-G are kind of like grouped together because they're all elongation factors. But like it's kind of like, oh, if you're gonna increase one of them, it's also a good idea to increase the other two.

But I'm hoping the model will learn that by itself. And then, yeah. But then we can also just like, also be like, oh, the model suggests that we add a whole bunch of creatine phosphate, but we're going to lower the creatine kinase. Then, we'll still do it, just so the model learns that that's a bad thing to do. But it's... It's probably not going to work. Yeah. So, when do we start putting our input?

So if you see that's not going to work, don't run that one. Or if you want to, if you want to change some of what the...

Okay. Okay, cool. Yeah. Okay. Yeah. Okay, that sounds great. The other thing that you can do as well, if you don't think it's going to run...

We'll sit down and work together. We'll actually classify that as a failure, even though we don't know if it is or not.

We'll classify it as a failure, we'll put it back into the dataset, we'll retrain another model, and then we'll come up with another set of solutions.

So you only kind of put your priors or your knowledge or biases into it.

Okay. Perfect. Okay. That's good. If we stay convinced of something, we can keep reminding it of that. Yeah. Okay. Yeah, because I don't want to just, like, bias it too much towards one data set. And so I was like, oh, like, for sometimes, like, I do want to just do it and just see what happens.

And if it fails, then we're like, okay, then maybe our intuition is good. Yeah, well, I guess we'll see what the model suggests. Okay, so there are some restrictions, unfortunately. So, like I said, the volume restrictions and the table to the right are going to be constant. So what we're going to change inside

Those components, or the mixtures, are the concentrations of each reagent. So that's like the overall pure reaction. And then so for like, now focusing on the protein mix, no protein should be zero, obviously. And then so the overall protein concentration should lie 8 to 25 mg per ml.

But honestly, I don't think that's a factor that we have to like, really think of, unless like the... Soft concern. It's like a soft concern. Because we can dilute it more or concentrate it more, depending on what the model suggests or what the AI suggests. But then the biggest one, I don't know how to do this for the...

Just linear constraints, right?

Let's say the model is like, oh, let's write 200 millimolar heaps. Then it actually increases the volume of heaps.

Okay, yeah, we can add constraints into that. Once I do that, I'll probably ask you to make sure I have a constraint.

I just wanted to flag one thing, which is just like, one of the knobs we don't want to turn but can turn is some of these salts. We can change the stock concentration. It's very much not ideal, but it's possible, and DMO has stronger opinions of which ones are and aren't changeable.

So basically, we can increase the head volume by increasing the top volume.

Some of these are optimized already.

Perfect. Yes, I will move it over there. Thank you, Yemo. That was a good instinct on the slides. And then, so what are previous publications? There's only actually one publication. But basically, like, um...

Yeah, I already said it's going to go in. It's been 14 minutes.

Two questions. Yes.

And I think that's basically the end of the PowerPoint. And then now, we're going to take a look at the template. So I put it already in the... Yeah, so basically if you

open it, this is kind of the template that we use to make our protein mix. So we have all our proteins and then the concentration we want in the protein mixture and then

Okay, it's frozen. Give me one sec. Okay, now it's not frozen. And then... Oh my gosh.

Okay, that's enough for tonight. Technical difficulties.

Okay, I'm gonna move slow. Yeah, so this is our basically our protein concentration, and then the model is going to suggest different protein concentrations that we want in our mixture, and then so we'll assemble it and then This one we're not going to use for now, because we're going to keep the volumes the same, so it doesn't matter. And then this is the template for our energy mix that we usually use.

This I think I'm providing you this basically so that you can kind of see the restraints depending on the volume. I'm not sure how to do it. There's definitely a formula. So let's just say like we increase this to like 200, then the volume, we only have 2.1 microliters.

left over.

I mean, we can increase that volume, but the ratios are going to stay the same. So, if that makes sense. I know how the formula should look like. Basically, if you give us a recipe that requires us to pipette 0.1 microliters, we'll just 10x the volume of energy mix we make, and then that's not the problem anymore. The problem is we can't add negative water. So as long as you cover that, then we're good.

The other thing that's worth noting is ATP, GTP, CTP, and UTP are master mixed at fixed 2-to-1 ratios, so you should treat that as just one component. But at the same time, I want to change that. Reasonable finding. Because that one's like, I feel like that is actually going to have a...

I apologize. Thank you. Yeah. Because I feel like ATP and GTP, we can increase it by a lot more and then see what the outcome Basically, once we put what the AI gives us, we're going to then... And then it just puts in exactly what the concentration... Oh, it's not even freaking loading. Wait one second.

Oh, does it really? Okay, maybe I should re... I'm gonna stop sharing and then re-share it again. Okay, now it's working. Oh, it's working? Now it's working. OK, so

basically then it outputs it into this table format of all the components and their concentrations and the final protein and the final pure reaction.

Is that kind of how you wanted it?

A dataset, I think, that I can look for.

I can do another one. Thank you for using your expertise.

Oh, that's a good... Like, you could highlight which variables you think are the most sensitive, and then he can just do, like, a well-informed random sampling therefrom, and that'd be the first set.

Yeah, let's try that. Because we're going to have protein mix hopefully by the end of this week. And then I can do exactly... It's not a picture of protein mix. And then I think we're gonna run that and we'll have that first data set. And that'll be like our, that'll be honestly our positive control or like.

The model will eventually learn those relationships, right? Yeah. Okay. Then why don't we, should we just try repeating the experiments from this paper that, of what they did? And then so I can just kind of get a grasp of like, oh, these are the most important...

Components and protein mix that should be tuned first. I mean I'll flag also that They don't really well sample small molecule mix that much and that's a really easy one for you to add extra samples of right so we can do like

We can do like the protein mixes of these, changing these factors. And I guess like you, like the, can we just be like, oh, can you come change these four components and then just give it a different like concentrations? Okay. And then maybe like, then we can also just do one, just to randomize or so while sampling for the energy mix.

Yeah. So I can, I'll make a list of like the proteins and then. For the energy mix, though, we'll just do a randomized. OK. And then if any of those look too crazy, just say you're changing it. Yeah. Yeah. Change it by one. Okay. Yeah. Yeah.

Okay. That sounds great. So yeah, so we'll have like kind of like a directional suggestion with these. And then, so yeah, so by the end of this week I hope we have like the baseline. And then. Next week, we can start trying what the AI suggests.

So, is there anything that you want from us by any chance?

I guess just like look over the material and then I'll provide the list of proteins and then the thing and then hopefully by the end of this week or early next week you'll have like the energy mix suggestions and the protein suggestions. Okay perfect.

Yes, I was trying to make everyone's lives a little easier.

So that's good.

If you have any questions about that, I mean, Yemo can describe it really well, but I feel pretty comfortable with it.

I'm not very confident describing things as constrained, so I might be a more tactfully useful person for that.

Dude, it's just linear combinations. It's so chill. It's like super duper chill. But yeah, I would love to talk about this. I think like this one will be the hardest thing for y'all to determine. I know how to do it. I think I got you. I used to do this in grad school a lot.

But yeah, yeah, I'll flag it if it starts being a revolution.

Yeah, thank you. Yeah, because I was like, when you guys asked me that question, I was like literally like, oh yeah, what is our goal, technically? Honestly, gut, I would have said 2x, just because I wanted to be conservative, but like, yeah, 5x is reasonable.

Right, because if they only changed like, you know, a few factors and they were able to achieve 5x, then like, you know. What can we achieve with AI?

Right.

And that sometimes takes a few more iterations to get to the ultimate solution. But you do see an increase in each iteration.

Yeah, that sounds good. How many iterations do you think it'll take until we're kind of like... That's the question we always get. That's the one that I always talk about.

Gotcha.

So, all I can say is once we have some experience and we've been here, I can say, based on empirical studies that I've done, the things about this.

The most efficient way for money is on your head.

Gotcha.

So, we'll try to do that. It's a very, very tough question to answer. That's something everybody asks.

Yeah.

Yeah, that makes sense. I feel like since we have a baseline and some studies that point us in the right direction, hopefully the mod will learn quickly. But I'm excited. So I guess like we'll test it out next week and then maybe we can like set a limit or like you know come up with a number that's like a good number of iterations we can do per week.

We tend to break it down to like, let's see if you want to take this one, let's go with that one.

So that way you're getting some intuition from yourself and your own expertise.

Right. Yeah, I think that's what I was planning to do, because you're going to suggest a protein mix, and if it looks good, then I was going to do a second iteration where I increase the ribosomes and the magnesium concentration to get the optima for that protein mix.

And then I'm hoping the model will eventually learn, if we were to always increase ribosomes, then you always have to increase magnesium of the energy mix. Yeah.

I really think it's one of those things where it's like, as a function, protein mix, like the ribosome only cares about those 36 things, all as one thing. I do think it kind of almost collapsed. We think, we think, this is our intuition. We'd love to be proven wrong, because that would be great for the video. I will say, so 20 of the proteins out of the 36 in the protein mix have one function, and it's literally to attach the amino acid to the tRNAs and then deliver it to the ribosome.

So, I don't... is there a way that, like, we can say, like, don't optimize this too much, because we don't want it to be optimized for the... for GFP. Yeah, which has a particular amino... One of the things I'm going to try to do...

For sure, that'd be awesome. Yeah.

We're about to do a laptop wipe for a project, but like immediately after that it'll be for sure. Okay, that timing is going to work out real nice. That timing is going to

work out real nice. Because often times, you run out of punctuation. Where do you change that? Where do you change this? Yeah, that'd be cool.

It's a cool front end. It was written by my roommate.

Okay, okay, okay.

Yeah.

Yeah, yeah, yeah. That's awesome. Okay, so let me move this over to the shared drive. And then let me see if I can just download this. And as far as setting expectations, making pure work is one of those long-term strategic goals. It's okay if it takes us a little time to get there.

The goal here is max success, but the fires we're putting out right now is getting our protein mix working at all, for instance. This kind of stuff is more like, over the years, we want to be doing a lot of optimizations like this. This is slightly more spacious than the other project.

We love going fast, but we're a little more patient with it.

So what is...

Well, ideally months for this, but it's not one of those things where it's like, by the end of the month we need to have five minutes. It's more like...

I think it's one of those things where it's like, I think it's one of those things where the reason why we're putting energy towards this project is because us blindly trying to optimize we're gonna... But if we can get this 5x improvement Then our product is more appealing. So it's one of those things where it's like it's useful for our community It's useful for us as a business But all of that is dependent on like getting our peer into tubes and into people's hands in the back

If we got the 5X before then, that would be amazing, but also we can't capitalize on it yet. Except internally, which would be awesome. We definitely love to be hitting these things before the end of the year, frankly. Honestly, at September, we have two big conferences where we're trying to go loud.

We'd love to be like, you know, be next pure optimized by find what matters 5x improvement on existing commercial providers That'd be really powerful

It would take us years.

I mean, like, unless we got you, like, two lab techs and the robot working, it would take you a very long time to do this just by, like, gumption, right? Like...

I mean, I would just, like, duplicate this and see if it works. And then repeat it. Instead of a 5x, we probably wouldn't have discovered it. And I have a question. Is RF, like relative fluorescence, the best measurement? Or is there another way you guys would?

We have a positive control always in our test cases, and then we just compare it to that. What we're learning is how...

Derivatives. What we can also do is have a standard curve always. But we have to validate that first, that it will stay consistent over the runs. But what that gives you is all of these numbers that get resolved to literal...

We like that. It's just a linear transform on the fluorescence data of the PLAN-GFP. We could give you those units and the raw data on the PLAN-GFP. The model would learn the exact same information. We can give you everything, too, if it's just, you know. I mean, usually the more information you get, the better. It's not more information, because one is just like a function on the other. So all you'd be learning is just the linear fit.

But the thing is, the positive control has a lot of variance. But let's just say if we had a standard for the highest fluorescence, then that would probably have less variance than the pure positive control. And it would be a better measure of just instrument noise.

That's an example. Let's start there. Maybe we should start. Yeah. Okay. We can throw a PLAM onto it. We can do a 27th column on Wednesday. I'm sorry, when I go to protein stocks, it says GFP?